CS-747 · Deep Learning for Computer Vision

Course Project — Problem Statements

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Cluster 1 — Visual Recognition, Verification & Representation Learning

These five problems all revolve around learning good image representations and comparing/classifying them — the throughline is embeddings and metric spaces rather than a fixed label set.

1. Few-Shot Fine-Grained Visual Recognition via Metric Learning

Build a few-shot recognition system for a fine-grained domain (e.g., bird species, plant diseases, or industrial defect types) where only a handful of labeled examples per class are available. Implement a prototypical-network-style metric learning approach that classifies a query image by its distance to class "prototypes" computed from a small support set, and compare against a standard fine-tuned CNN baseline trained on the same few examples.

Suggested dataset(s): CUB-200-2011 (birds) or a custom small fine-grained dataset (e.g., a PlantVillage subset).

Reference: *Snell, J., Swersky, K., & Zemel, R. (2017). Prototypical Networks for Few-shot Learning. NeurIPS.*

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2. Self-Supervised Contrastive Pretraining for Image Classification

Pretrain a CNN or small ViT backbone using a self-supervised contrastive/self-distillation objective on unlabeled images, then evaluate the learned representation by fine-tuning a linear classifier on a small labeled subset. Compare the label-efficiency of the self-supervised representation against a network trained from scratch with the same amount of labeled data.

Suggested dataset(s): STL-10 (designed for exactly this unlabeled + small-labeled setup) or CIFAR-100.

Reference: *Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., & Joulin, A. (2021). Emerging Properties in Self-Supervised Vision Transformers (DINO). ICCV.*

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3. Open-Set Recognition and Out-of-Distribution Detection

Train a closed-set image classifier and evaluate how well simple post-hoc scores (max softmax probability, max logit score, energy score) separate in-distribution test images from genuinely novel, unseen-class images at test time. Extend the analysis with at least one dedicated OSR/OOD scoring method and report AUROC/FPR95 rather than accuracy alone.

Suggested dataset(s): CIFAR-10/CIFAR-100 as in-distribution vs. SVHN, or a held-out subset of classes, as out-of-distribution.

Reference: *Vaze, S., Han, K., Vedaldi, A., & Zisserman, A. (2022). Open-Set Recognition: A Good Closed-Set Classifier is All You Need? ICLR.*

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4. Cross-Domain Person Re-Identification (Verification)

Train a Siamese/triplet-loss embedding network for person re-identification on one camera/dataset domain, then measure the accuracy drop when it is evaluated on a different, unseen domain (different cameras, lighting, or city). Implement at least one domain-generalization technique and quantify how much of the gap it recovers.

Suggested dataset(s): Market-1501 (source domain) and DukeMTMC-reID or MSMT17 (target domain).

Reference: *Nie, R., Ding, J., Zhou, X., & Li, X. (2024). Rethinking Normalization Layers for Domain Generalizable Person Re-identification. ECCV.*

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5. Vision Transformers vs. CNNs: A Robustness Study

Train a ResNet-style CNN and a Vision Transformer of comparable parameter count on the same classification dataset, then systematically compare their robustness to common corruptions (blur, noise, occlusion) and to a simple adversarial attack, alongside clean accuracy. The project should isolate whether architecture itself — not just parameter count or training recipe — drives any robustness gap.

Suggested dataset(s): CIFAR-100 or an ImageNet subset, evaluated with CIFAR-100-C / ImageNet-C style corruptions.

Reference: *Dosovitskiy, A., et al. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR.*

Cluster 2 — Object Detection

All five problems use detection as the task, each stressing a different real-world constraint (scale, speed, architecture, environment, or vocabulary) so groups cannot simply reuse one another's pipelines end to end.

6. Small Object Detection in Aerial/Drone Imagery

Adapt a modern one-stage detector to reliably detect very small objects (vehicles, people) in high-altitude drone or satellite images, where objects can occupy only a few pixels. Investigate at least one small-object-specific technique such as tiling/slicing inference, feature-pyramid enhancement, or attention-guided multi-scale fusion, and report performance specifically on the small-object size bucket.

Suggested dataset(s): VisDrone or DOTA.

Reference: *Zhang, Y., Ye, M., Zhu, G., Liu, Y., Guo, P., & Yan, J. (2024). FFCA-YOLO for Small Object Detection in Remote Sensing Images. IEEE Transactions on Geoscience and Remote Sensing.*

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7. Real-Time Object Detection on Edge Devices

Deploy a state-of-the-art real-time detector on a resource-constrained target (a laptop CPU, Raspberry Pi, or phone via a mobile export) and quantify the accuracy/latency/model-size trade-off after applying at least one compression technique (pruning, quantization, or distillation). Produce a clear speed-vs-accuracy curve, not just a single working demo.

Suggested dataset(s): COCO, or a smaller custom subset for faster iteration.

Reference: *Tian, Y., Ye, Q., & Doermann, D. (2025). YOLOv12: Attention-Centric Real-Time Object Detectors. NeurIPS.*

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8. Transformer-Based End-to-End Object Detection

Implement or fine-tune a DETR-style detector that treats detection as a set-prediction problem (no anchors, no NMS) and compare its training convergence, accuracy, and failure modes against a classic anchor-based detector on the same dataset. Investigate why transformer detectors are historically slower to converge, and evaluate at least one fix (e.g., a deformable-attention or denoising-query variant).

Suggested dataset(s): COCO, or PASCAL VOC for lower compute budgets.

Reference: *Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., & Zagoruyko, S. (2020). End-to-End Object Detection with Transformers (DETR). ECCV.*

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9. Object Detection Under Adverse Weather and Low Light

Build a detection pipeline that stays accurate in fog, rain, or nighttime conditions, comparing (a) a detector trained only on clean images, (b) one trained with weather-augmented data, and (c) a two-stage enhance-then-detect pipeline. Evaluate specifically on a real (not just synthetically augmented) adverse-condition benchmark to test genuine generalization.

Suggested dataset(s): BDD100K (has weather/time-of-day labels) or DAWN.

Reference: *Tiwari, A. K., Pattanaik, M., & Sharma, G. K. (2024). Low-light DEtection TRansformer (LDETR): Object Detection in Low-Light and Adverse Weather Conditions. Multimedia Tools and Applications.*

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10. Open-Vocabulary Object Detection

Build a detector that can localize object categories never seen during training by leveraging a vision-language embedding space (e.g., CLIP-style text embeddings as classifier weights instead of a fixed label set), and measure zero-shot detection accuracy on held-out novel categories versus base categories. Report base-class and novel-class performance separately rather than one blended number.

Suggested dataset(s): LVIS (naturally long-tailed, suited to a base/novel split) or a custom base/novel split of COCO.

Reference: *Xi, X., Huang, Y., Luo, R., & Qiu, Y. (2025). OW-OVD: Unified Open World and Open Vocabulary Object Detection. CVPR.*

Cluster 3 — Semantic, Instance & Panoptic Segmentation

All five problems produce a pixel-level output, but each pushes on a different constraint: label scarcity, real-time speed, unifying stuff/things, or a promptable foundation model.

11. Medical Image Segmentation with U-Net Variants

Segment an anatomical structure or lesion (e.g., organs on CT, or a tumor on MRI) using a U-Net baseline, then implement a transformer-augmented variant that replaces or supplements the encoder with self-attention. Compare Dice/Hausdorff scores and, importantly, data efficiency when only a small labeled training set is available — a realistic constraint in medical imaging.

Suggested dataset(s): Synapse multi-organ CT dataset, or the ISIC skin lesion dataset.

Reference: *Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A. L., & Zhou, Y. (2021). TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation. arXiv:2102.04306 (extended in Medical Image Analysis, 2024).*

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12. Panoptic Segmentation for Scene Understanding

Build a system that jointly produces “stuff” (sky, road, grass) and “thing” (car, person) segmentation in a single unified output, reconciling both predictions into one consistent panoptic map. Analyze where and why conflicts arise between the semantic and instance branches, and implement a fusion heuristic (or a unified query-based head) to resolve them.

Suggested dataset(s): Cityscapes (panoptic annotations) or COCO-Panoptic.

Reference: *Kirillov, A., He, K., Girshick, R., Rother, C., & Dollár, P. (2019). Panoptic Segmentation. CVPR.*

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13. Real-Time Semantic Segmentation for Autonomous Driving

Design or adapt a lightweight segmentation network that runs at real-time frame rates on driving-scene imagery (e.g., drivable area and lane segmentation), explicitly trading off encoder depth/resolution for speed. Benchmark actual FPS on a single GPU/CPU alongside mIoU rather than reporting accuracy alone.

Suggested dataset(s): Cityscapes or BDD100K (drivable area / lane labels).

Reference: *Che, Q. H., Le, D. T., Pham, M. Q., Nguyen, V. T., & Lam, D. K. (2024). TwinLiteNetPlus: A Real-Time Multi-Task Segmentation Model for Autonomous Driving. arXiv:2403.16958.*

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14. Weakly-Supervised Semantic Segmentation

Train a segmentation model using only image-level class labels (no pixel masks) by generating pseudo-masks from class activation maps and refining them — directly building on the CAM techniques from this course's visualization unit. Quantify the accuracy gap versus a fully-supervised model trained on the same images with full pixel annotations.

Suggested dataset(s): PASCAL VOC 2012 (the standard WSSS benchmark).

Reference: *Lin, C.-S., Wang, C.-Y., Wang, Y.-C. F., & Chen, M.-H. (2025). SemPLeS: Semantic Prompt Learning for Weakly-Supervised Semantic Segmentation. WACV.*

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15. Promptable Segmentation with Segment Anything (SAM)

Use a pretrained Segment Anything Model as a zero-shot segmenter on a domain it was not specifically trained for (e.g., satellite imagery, microscopy, or a custom everyday object category). Evaluate how

prompt type (point, box, or automatic mask generation) affects mask quality, and implement a lightweight adapter or fine-tuning step to close any domain gap found.

Suggested dataset(s): Any domain-specific image set with ground-truth masks under-represented in SAM's original SA-1B training data (e.g., remote-sensing or medical images).

Reference: *Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., et al. (2023). Segment Anything. ICCV.*

Cluster 4 — Explainability, Visualization & Trust

This cluster extends the course's CAM/Grad-CAM unit directly: every problem asks "why did the network decide this," evaluated quantitatively rather than by eyeballing heatmaps.

16. Comparative Study: Grad-CAM vs. Grad-CAM++ vs. Integrated Gradients

Implement all three attribution methods on the same trained CNN classifier and design a quantitative comparison — not just visual inspection — using a metric such as deletion/insertion curves or pointing-game accuracy against object bounding boxes, to determine which method most faithfully reflects what the network actually used to decide.

Suggested dataset(s): An ImageNet validation subset with bounding-box annotations, or PASCAL VOC.

Reference: *Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV.*

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17. Concept-Based Explainability (TCAV)

Go beyond pixel-level saliency by testing whether a trained classifier's predictions are sensitive to human-interpretable concepts (e.g., "stripes," "wheels," "redness") that were never explicitly labeled as classes, using directional derivatives in the network's activation space. Use the technique to audit a model for at least one potentially spurious concept-class association.

Suggested dataset(s): A subset of ImageNet plus a small set of user-collected concept example images (e.g., "striped" textures).

Reference: *Kim, B., Wattenberg, M., Gilmer, J., Cai, C., Wexler, J., Viegas, F., & Sayres, R. (2018). Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV). ICML.*

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18. Robustness of Saliency Explanations Under Adversarial Perturbation

Demonstrate and quantify how fragile standard saliency-based explanations can be by crafting small, prediction-preserving perturbations that drastically change a Grad-CAM/saliency map without changing the model's output class. Evaluate whether a more robust attribution method resists this manipulation better.

Suggested dataset(s): Any standard classification dataset (e.g., CIFAR-10 or a small ImageNet subset) — the focus is the attack/defense on the explanation, not the base classifier.

Reference: *Ghorbani, A., Abid, A., & Zou, J. (2019). Interpretation of Neural Networks is Fragile. AAAI.*

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19. Explainability for Vision Transformers

Compare naive attention-map visualization, attention rollout, and a gradient-aware relevance-propagation method for explaining a Vision Transformer's classification decisions, evaluating each with the same deletion/insertion or pointing-game protocol used for CNN attribution methods so the comparison across architectures is apples-to-apples.

Suggested dataset(s): An ImageNet validation subset with bounding boxes, or PASCAL VOC.

Reference: *Chefer, H., Gur, S., & Wolf, L. (2021). Transformer Interpretability Beyond Attention Visualization. CVPR.*

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20. Uncertainty Quantification in Deep Vision Models

Extend a standard classification or segmentation network to output calibrated uncertainty alongside its prediction, implementing both aleatoric (data noise) and epistemic (model “I don't know”) uncertainty estimation. Show on a held-out set that the model's high-uncertainty predictions correlate with its actual errors.

Suggested dataset(s): CIFAR-10/100 for classification, or Cityscapes for segmentation uncertainty.

Reference: *Kendall, A., & Gal, Y. (2017). What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? NeurIPS.*

Cluster 5 — Deep Generative Models & Applications

This cluster covers the course's generative-models unit, from the core GAN-vs-diffusion question through the classic image-editing applications named in the syllabus.

21. Diffusion Models vs. GANs for Image Generation

Train a GAN and a diffusion model of comparable capacity on the same image dataset, then compare sample quality (FID), training stability, and mode coverage/diversity between the two paradigms — giving a concrete, evidence-based answer to "which is actually better, and under what conditions" rather than a purely qualitative comparison.

Suggested dataset(s): CIFAR-10 or CelebA (small enough to train both model families from scratch within course time constraints).

Reference: *Dhariwal, P., & Nichol, A. (2021). Diffusion Models Beat GANs on Image Synthesis. NeurIPS.*

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22. Deep Generative Image Inpainting

Use a pretrained diffusion model to fill in masked/missing regions of an image without task-specific retraining, exploring how mask shape/size and the number of resampling steps affect realism and semantic consistency of the completed region, benchmarked against a GAN-based inpainting baseline.

Suggested dataset(s): CelebA-HQ or Places2 (standard inpainting benchmarks).

Reference: *Lugmayr, A., Danelljan, M., Romero, A., Yu, F., Timofte, R., & Van Gool, L. (2022). RePaint: Inpainting Using Denoising Diffusion Probabilistic Models. CVPR.*

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23. Image Super-Resolution via Generative Models

Build a single-image super-resolution pipeline using a diffusion-based method, comparing perceptual quality and inference speed against a GAN-based super-resolution baseline, with particular attention to the number of denoising steps needed for acceptable quality — diffusion models' main practical weakness.

Suggested dataset(s): DIV2K (the standard super-resolution benchmark).

Reference: *Yue, Z., Wang, J., & Loy, C. C. (2023). ResShift: Efficient Diffusion Model for Image Super-Resolution by Residual Shifting. NeurIPS.*

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24. Unpaired Image-to-Image Translation with CycleGAN

Implement CycleGAN to learn a mapping between two visual domains without any paired training examples (e.g., photo-to-painting, summer-to-winter, or healthy-to-diseased medical images), and specifically investigate the cycle-consistency loss's role by ablating it and showing what fails without it.

Suggested dataset(s): Any standard CycleGAN dataset (horse2zebra, summer2winter, monet2photo) or a domain pair of the student's choice.

Reference: *Zhu, J.-Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. ICCV.*

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25. Deepfake Generation and Detection

Generate face-swapped or reenacted images/video clips using an existing generative pipeline, then build a detector for this content. Critically evaluate the detector against genuinely in-the-wild deepfakes rather

than only the same generator's outputs, since recent evidence shows detectors trained on lab-generated data generalize poorly to real-world manipulated media.

Suggested dataset(s): FaceForensics++ for training, evaluated additionally against a small hand-collected or public in-the-wild sample if feasible.

Reference: *Chandra, N. A., Lee, H., Murtfeldt, R., Qiu, L., Karmakar, A., et al. (2025). Deepfake-Eval-2024: A Multi-Modal In-the-Wild Benchmark of Deepfakes Circulated in 2024. arXiv:2503.02857.*

Cluster 6 — Emerging & Advanced Vision Research

These five problems sit just beyond the syllabus's explicit list but grow directly out of it, giving strong groups room to engage with genuinely current (2023–2026) research for CO4.

26. Neural 3D Scene Reconstruction (NeRF / Gaussian Splatting)

Reconstruct a 3D scene from a set of 2D photos taken around an object or room and render novel viewpoints, comparing an implicit neural radiance field approach against the newer explicit 3D Gaussian Splatting approach on training time, rendering speed, and visual quality.

Suggested dataset(s): A small self-captured multi-view photo set (a phone camera is sufficient) or a standard benchmark scene (e.g., NeRF's “Lego” scene or a Mip-NeRF360 scene).

Reference: *Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (2023). 3D Gaussian Splatting for Real-Time Radiance Field Rendering. ACM Transactions on Graphics (SIGGRAPH).*

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27. Vision-Language Models for Zero-Shot Recognition

Use a pretrained vision-language embedding model to perform zero-shot image classification and image/text retrieval without task-specific fine-tuning, and probe its failure modes systematically (fine-grained categories, counting, spatial relations, or unusual class names) to characterize exactly where “zero-shot” breaks down.

Suggested dataset(s): A zero-shot classification benchmark such as an ImageNet subset, plus a fine-grained dataset (e.g., Oxford-IIIT Pets) to surface failure cases.

Reference: *Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning Transferable Visual Models From Natural Language Supervision (CLIP). ICML.*

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28. Spatio-Temporal Deep Learning for Video Action Recognition

Build a video action-recognition system using a video transformer that jointly models spatial and temporal attention, and compare it against a 3D-CNN baseline on accuracy, training cost, and the amount of video needed for reasonable performance, since transformers are known to be more data-hungry.

Suggested dataset(s): UCF-101, or a subset of Kinetics-400 for compute feasibility.

Reference: *Arnab, A., Dehghani, M., Heigold, G., Sun, C., Lučić, M., & Schmid, C. (2021). ViViT: A Video Vision Transformer. ICCV.*

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29. Efficient Hybrid CNN-Transformer Architectures for Mobile Vision

Design or adapt a hybrid architecture (combining convolution's efficiency with attention's global context, optionally including newer state-space/Mamba-style mixing) for image classification, targeting a specific latency budget on CPU or mobile hardware. Report the full accuracy-vs-latency-vs-parameter-count trade-off against both a pure CNN and a pure ViT baseline.

Suggested dataset(s): CIFAR-100, or a mobile-scale ImageNet subset.

Reference: *Hatamizadeh, A., & Kautz, J. (2025). MambaVision: A Hybrid Mamba-Transformer Vision Backbone. CVPR.*

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30. Multimodal Foundation Models for Visual Question Answering

Build a VQA system by connecting a frozen pretrained vision encoder to a frozen (or lightly fine-tuned) language model through a small trainable bridging module. Evaluate both answer accuracy and, importantly, whether the model's answers are actually grounded in the image — via a targeted probing set of questions with visually verifiable answers — rather than exploiting language priors.

Suggested dataset(s): VQA v2, or a smaller VQA benchmark subset for course-scale compute.

Reference: *Li, J., Li, D., Savarese, S., & Hoi, S. (2023). BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. ICML.*